

recommend the recent book by Gelfand, Kapranov & Zelevinsky [230] for study.

Our definitions (starting with the Definition 9.1 of π -induced subdivisions) are different in appearance from, but equivalent to, the original set-up by Billera & Sturmfels [78], which works with vertex sets rather than polytopes. Only the explicit version of π -coherent subdivisions in Definition 9.2 might be new here. We also refer to the survey of fiber polytopes in [534], and to the alternative set-up (via normal fans) in [79].

Similarly, the original definition of secondary polytopes by Gel'fand, Zelevinsky & Kapranov [231, 232] (see [230, Ch. 7]!) was quite different from the one, due to Billera & Sturmfels [78], which we have presented in Definition 9.9. Our presentation also reverses the historical order of things: the ingenious construction of [231] now appears as a very special case of the fiber polytope construction; let us just say that the secondary polytopes are in many respects the most fundamental case. In particular, every fiber polytope can be written as a projection of a secondary polytope; see Exercise 9.6.

The “permuto-associahedron” $K\Pi_{n-1}$ is a combinatorial object introduced by Kapranov [313] (he denotes it as KP_n). The construction as a polytope, and the generalization to “Coxeter-associahedra,” by Reiner & Ziegler [453], were born in December 1992, so to speak (and not baptized). The nonlinear effects appearing in this, like the sphericity in Proposition 9.12, suggest that there is much more to be discovered and that the constructions are not yet well understood.

Problems and Exercises

9.0 For which of the examples of polytopes discussed in Lecture 0 can we, by now, give complete combinatorial descriptions?

Which of them are related by projections?

Which can we represent as fiber polytopes associated with a projection of simpler polytopes?

9.1 Compute the fiber polytope for the projection in Example 9.3. For this, compute both the vertex coordinates, corresponding to the six coherent tight sections.

(According to our discussion, you should get a hexagon!)

Also, compute the point in $\mathbb{R}^3$ which corresponds to the noncoherent tight section of Example 9.3. Does it lie in the relative interior of the fiber polytope?

Finally, use the methods described in this chapter to derive a description of the fiber polytope by equations and inequalities.

9.2 For the projection

$$\pi : C_3^\Delta \longrightarrow [-2, +2], \quad \mathbf{x} \longmapsto 2x_1 + x_2,$$

enumerate *all* the π -induced subdivisions, and identify the π -coherent ones among them. Draw the whole poset, and show how it “retracts” to the subposet of π -coherent subdivisions.

Compute the fiber polytope, and describe the position in the fiber polytope of the points that correspond to noncoherent subdivisions.

- 9.3 For a polytope projection $\pi : P \longrightarrow Q$, let $\mathcal{F}$ be any π -induced subdivision. Show that for every linear function $\mathbf{c} \in (\mathbb{R}^P)^*$ there is a *relative coherent subdivision* $\mathcal{G}^{\mathbf{c}}$ with $\mathcal{G}^{\mathbf{c}} \leq \mathcal{F}$.
(The construction can be done analogously to Definition 9.2.)

Show that if $\mathbf{c}$ is generic, then $\mathcal{G}^{\mathbf{c}}$ is tight. Conclude that the minimal elements of $\omega(P, Q)$ are exactly the tight subdivisions.

- 9.4 Given a monomial $m = x_1^{t_1} x_2^{t_2} \cdots x_d^{t_d}$, define its *degree* by

$$\deg(m) := \begin{pmatrix} t_1 \\ \vdots \\ t_d \end{pmatrix} \in \mathbb{R}^d.$$

For a polynomial in d variables,

$$f = \sum_i \alpha_i m_i \in \mathbb{R}[x_1, \dots, x_d],$$

for $\alpha_i \in \mathbb{R}$ and monomials m_i , define its *Newton polytope* by

$$\text{Newton}(f) := \text{conv}\{\deg(m_i) : \alpha_i \neq 0\}.$$

- (i) Show that $\text{Newton}(f \cdot g) = \text{Newton}(f) + \text{Newton}(g)$. From this, describe the Newton polytope $\text{Newton}(f^k)$. What can you say about $\text{Newton}(f+g)$, and about $\text{Newton}(f+\varepsilon g)$ for small enough $\varepsilon > 0$?
 - (ii) Describe $\text{Newton}(\det(x_{ij}))$, for the determinant of an $(n \times n)$ -matrix, with $d = n^2$ different variables as entries.
- 9.5 If $\pi : P \longrightarrow Q$ is a projection of polytopes, then the π -induced subdivisions of Q only use vertices in the finite set $\pi(\text{vert}(P))$, which need not all be vertices of Q .
- Show that if $P = \Delta_p$ is a simplex, then all the regular π -induced subdivisions of Q are π -coherent.

- 9.6 Let $\pi : P \longrightarrow Q$ be a projection of polytopes, where $P \subseteq \mathbb{R}^p$ is a polytope on n vertices. Show that the fiber polytope $\Sigma(P, Q)$ can be constructed by projecting the secondary polytope of P :

$$\Sigma(P, Q) = \pi(\Sigma(P)).$$

(Billera & Sturmfels [78])

- 9.7 Verify the formulas for the volume (area) and the barycenter of the $(n+1)$ -gon

$$C_2(n+1) = \text{conv}\left\{\binom{i}{i^2} : i = 0, 1, 2, \dots, n\right\}$$

What about the volume and the barycenter of the general cyclic polytopes $C_d(n+1)$?

- 9.8 Let P and Q be polygons in the plane. Show that the fiber polytope of the projection

$$P \times Q \longrightarrow P + Q$$

from the product to the Minkowski sum has a fiber polytope that is isomorphic to $P + (-Q)$.

What goes wrong here if Q degenerates to a line segment?

- 9.9 Define the *join* $P * Q$ of two polytopes to be the convex hull of P and Q , if they are placed into affine subspaces of some $\mathbb{R}^d$ such that their affine hulls $\text{aff}(P)$ and $\text{aff}(Q)$ are skew.

Show that $P * Q$ is a polytope of dimension $\dim(P) + \dim(Q) + 1$, and that up to affine equivalence the join does not depend on the choice of affine subspaces.

Show that the secondary polytope $\Sigma(P * Q)$ of the join is isomorphic to $\Sigma(P) \times \Sigma(Q)$.

(Dalbec [170])

- 9.10 Consider the projection $\pi : \mathbb{R}^n \longrightarrow \mathbb{R}$, $\Delta_{n-1} \longrightarrow [1, n]$ given by $\mathbf{e}_i \longmapsto i$.

Show that the fiber polytope $\Sigma(\Delta_{n-1}, [1, n])$ of this map is combinatorially equivalent to a $(n-2)$ -cube. Is it in fact affinely isomorphic to C_{n-2} ?

(Gelfand, Kapranov & Zelevinsky [230, Example 7.3.A].)

- 9.11 Compute the secondary polytope of $\Delta_{n-1} \times \Delta_1$.

For that first determine the dimension of the secondary polytope, then determine its set of vertices. Before you start to actually compute vertices, you should figure out the combinatorics of the polytope you get. (It is a good old friend!)

Also, study the secondary polytopes of $\Delta_{n-1} \times \Delta_{m-1}$.

(See Gelfand, Kapranov & Zelevinsky [230, Examples 7.3.C,D]; The secondary polytopes of products of simplices are really complicated: see Babson & Billera [32]!)

- 9.12 Show that if Z is a d -zonotope with n zones, then the fiber polytope of the canonical projection

$$\pi : C_n \longrightarrow Z$$

is a zonotope as well.

(Billera & Sturmfels [78, Thm. 6.1])

Compute the fiber polytopes for the projections

$$C_4 \longrightarrow P_2^8, \quad C_5 \longrightarrow P_2^{10}, \quad \text{and} \quad C_6 \longrightarrow P_2^{12},$$

where P_2^{2i} denotes a centrally symmetric $2i$ -gon. For the projection $C_6 \longrightarrow P_2^{12}$, the answer depends on the specific choice of a 12-gon. Determine the (five) different f -vectors that occur in this case. (Sturmfels [533])

- 9.13 If P and Q are disjoint polytopes in $\mathbb{R}^d$, show that the region between them can be triangulated without new vertices. That is, there exists a simplicial subdivision of $\text{conv}(P \cup Q)$ whose vertex set is $\text{vert}(P) \cup \text{vert}(Q)$, and such that there are subcomplexes that triangulate P and Q .

(Hint: Start with a convex function that is linear on P and constant on Q , and use it to “lift” Q . Then take the convex hull, and perturb all the vertices. The result is due to Goodman & Pach [233].)

- 9.14 For arbitrary polytopes $P \subseteq \mathbb{R}^p$ and $Q_1, Q_2 \subseteq \mathbb{R}^q$, show that

$$\Phi(P, Q_1 + Q_2) \supseteq \Phi(P, Q_1) + \Phi(P, Q_2).$$

Show that equality does not hold in general.

(Rambau & Ziegler [450])

- 9.15 Compute the mapping polytopes

$$\Phi((\Delta_p, \frac{1}{p}\mathbf{1}), (\Delta_q, \frac{1}{q}\mathbf{1})),$$

and describe them combinatorially.

- 9.16* What is a *cofiber polytope*?

(This should be a polytope that is naturally associated to any inclusion of polytopes $P \hookrightarrow Q$.)

References

- [1] HANS ACHATZ & PETER KLEINSCHMIDT: *Reconstructing a simple polytope from its graph*, in: “Polytopes — Combinatorics and Computation” (G. Kalai & G.M. Ziegler, eds.), *DMV-Seminars* **29**, Birkhäuser Basel 2000, 155–165. (95)
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- [3] RON M. ADIN: *A new cubical h -vector*, *Discrete Math.* **157** (1996), 3–14. (280)
- [4] RON M. ADIN: *On h -vectors and symmetry*, in: “Jerusalem Combinatorics ’93” (H. Barcelo and G. Kalai, eds.), *Contemporary Mathematics* **178**, Amer. Math. Soc. 1994, 1–20. (280)
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- [6] OSWIN AICHHOLZER: *Extremal properties of 0/1-polytopes of dimension 5*, in: “Polytopes — Combinatorics and Computation” (G. Kalai & G.M. Ziegler, eds.), *DMV-Seminars* **29**, Birkhäuser Basel 2000, 111–130. (26)

The numbers at the end of each reference indicate the pages on which it occurs.